INCH-POUND
MIL-DTL-24643/63B
27 November 2023
SUPERSEDING
MIL-DTL-24643/63A(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/63A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSYSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable type is specifically designed for single-circuit (see figure 1), three-phase, wye connected (with neutral wire), 60- or 400-hertz, electrical power distribution. The designation Y represents the wye connection. Using two conductors per phase reduces the series inductance to 50 percent of the single conductor per phase value. In addition, the symmetrical spacing of the phase conductors from the neutral wire provides for balanced voltage drop when three-phase loads are applied.

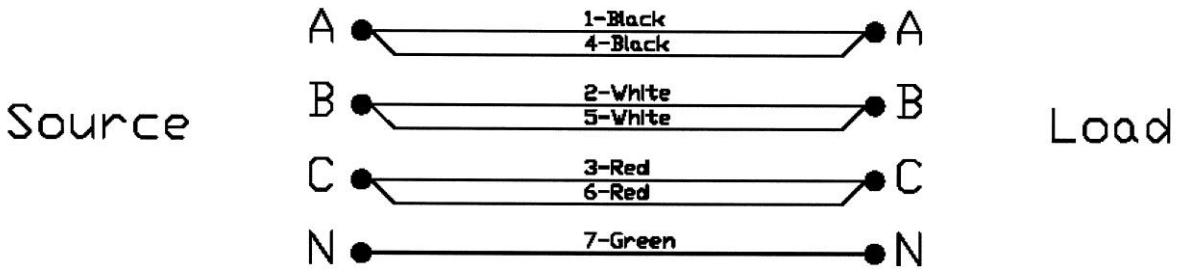


FIGURE 1. LSYSGU single-circuit user diagram.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated, ASTM B8 (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free stripping insulation.
Third	-	Thermoset insulation, extruded or taped (see table I for wall thickness).
Fourth	-	An optional glass braid.

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Fifth - An optional covering (unless required by 3.3.6 of MIL-DTL-24643).

Sixth - The seven conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. The center conductor shall be the green neutral (N), with the six phase conductors around it in the sequence A, B, C, A, B, C. Fillers may be used as necessary to form a firm, well-rounded assembly. Conductor identification code applied by method 1 per table II (and figure 1) conductor number and color; or method 3 per table II colors; or letter identification code applied by method 5 per table II identification code. If method 1 is used, outer surface to be printed shall appear black.

Seventh - An optional binder.

Eighth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. LSYSGU sizes and characteristics.

Military part no. M24643/63	Type size	Conductor			Cable					
		AWG	Stranding (inch)	Insulation thickness min. avg. (inch)	Amps per phase (2 conductors)		Conductor resistance (ohms)	Insulation resistance (megohms)	Jacket thickness abs. min. (inch)	Overall diameter max. (inch)
					40 °C ambient	50 °C ambient				
-01UN	3	16	7 x 0.0192	0.018	18	16	4.300	500	0.040	0.535
-02UN	4	14	7 x 0.0242	0.018	29	27	2.680	500	0.040	0.580
-03UN	6	12	7 x 0.0305	0.018	39	36	1.720	500	0.040	0.640
-04UN	9	10	7 x 0.0385	0.018	62	58	1.080	500	0.040	0.730
-05UN	16	8	7 x 0.0486	0.018	84	77	0.679	500	0.050	1.000
-06UN	40	4	19 x 0.0469	0.030	145	133	0.290	215	0.050	1.206
-07UN	60	2	37 x 0.0424	0.035	193	178	0.190	190	0.060	1.426
-08UN	75	1	37 x 0.0476	0.035	237	218	0.134	175	0.060	1.541
-09UN	100	0	61 x 0.0416	0.035	272	254	0.106	160	0.060	1.669
-10UN	125	00	61 x 0.0467	0.040	320	294	0.0888	150	0.065	1.828
-11UN	150	000	61 x 0.0524	0.040	376	346	0.0674	135	0.065	1.990

TABLE II. LSYSGU identification code and color.

Conductor no.	Identification code	Color
1	A	Black
2	B	White
3	C	Red
4	A	Black
5	B	White
6	C	Red
7	N	Green

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (root mean square, min.)	
AWG sizes 16 through 10	3,000 VAC
AWG sizes 8 and larger	5,000 VAC
Insulation resistance (megohms/1,000 feet, min.)	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (scrapes, min.) (type sizes 40 [8 AWG] and smaller) (extruded insulation only)	250
Crack resistance sizes 3 through 100 (applicable to constructions using optional covering)	No damage

Group B:

Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160

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Tear (lb/in thickness, min.)	35
Cold bending (cable)	No failure

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation – method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Cable fill (sealant) removability	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [20±2 °C])	No damage
Accelerated service (1.8 times the amps per phase at 122 °F [50 °C] ambient)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Gas flame (circuit integrity) (3 hours)	No failure
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSYSGU (all sizes)	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-079)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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